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Coupler unit for work machine

Abstract

A coupler unit for coupling a work tool with a work machine includes a coupling bracket. The coupler unit includes a pin, at least one side plate, a boss extending from the side plate, and a slider movably coupled with the boss. Based on a contact of the work tool with the slider, the slider is configured to move relative to the boss from a first position to a second position. In the first position, the slider is misaligned relative to the boss and the side plate, and the slider prevents receipt of the pin within the boss and the side plate. In the second position, each of the coupling bracket, the slider, the boss, and the side plate are in alignment with each other, and the pin is receivable within each of the coupling bracket, the slider, the boss, and the side plate.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to a coupler unit for a work machine, and more particularly, to the coupler unit for coupling a work tool with the work machine.

BACKGROUND

(2) A work machine, such as, wheel loader typically includes a coupler unit. The coupler unit allows coupling of different work tools (or implements), such as, a bucket, a scooper, a blade, a backhoe, etc., with the work machine. Various coupler units and mechanisms have been devised to facilitate quick interchange of the work tools so as to enable use of the work machine for different work functions. The coupler unit includes a locking pin that allow a quick connection of the coupler unit with different work tools. The locking pin may be received within a bore of the work tool so as to couple the coupler unit with the work tool.

(3) For the work tool to engage with the locking pin, the bore of the work tool needs to be in alignment with the locking pin. However, in some cases, the work tool may not be positioned appropriately relative to the coupler unit due to which the bore of the work tool may be misaligned in relation to the locking pin. In such cases, the locking pin may fully extend without engaging with the work tool. Generally, the coupler unit includes an indication system that generates an alert or indication when the locking pin has fully extended to notify an operator that the work tool is

coupled with the coupler unit. However, current indication systems do not consider an engagement of the locking pin with the work tool. Thus, it may be likely that the locking pin may fully extend and trigger the indication system even when the locking pin is not engaged with the work tool, thereby providing a false coupling indication.

(4) Overall, a misalignment of the work tool and the coupler unit may lead to a delay in coupling of work tools and may also increase a probability of false coupling indications by the indication system. Such false coupling indications may be misleading and may impact a reliability and usability of the coupler units, which may not be desirable. In some cases, the operator may have to get down from an operator cabin of the work machine and visually check whether the work tool is in alignment with the locking pin, which may increase downtime and may also involve additional operator efforts.

(5) U.S. Patent Application 2017/0107699 describes a coupler unit for coupling a work tool to a machine. The coupler unit comprises an indicator coupled to the coupler unit and pivotable between a first position and a second position. The second position is indicative of the work tool coupled to the machine. A coupling member is movable by an actuator to couple the work tool to the machine. The coupling of the work tool by the coupling member causes the indicator to pivot from the first position to the second position.

SUMMARY OF THE DISCLOSURE

(6) In an aspect of the present disclosure, a coupler unit for coupling a work tool with a work machine is provided. The work tool includes a coupling bracket. The coupler unit includes a pin configured to couple with the coupling bracket of the work tool. The coupler unit also includes at least one side plate. The coupler unit further includes a boss extending from the side plate. The coupler unit includes a slider movably coupled with the boss. Based on a contact of the work tool with the slider, the slider is configured to move relative to the boss from a first position to a second position. In the first position, the slider is misaligned relative to the boss and the side plate, and the slider prevents receipt of the pin within the boss and the side plate. In the second position, each of the coupling bracket, the slider, the boss, and the side plate are in alignment with each other, and the pin is receivable within each of the coupling bracket, the slider, the boss, and the side plate.

(7) In another aspect of the present disclosure, a work machine is provided. The work machine includes a frame. The work machine also includes a linkage assembly coupled to the frame. The linkage assembly includes a work tool including a coupling bracket. The linkage assembly also includes a coupler unit configured to couple with the work tool. The coupler unit includes a pin configured to couple with the coupling bracket of the work tool. The coupler unit also includes at least one side plate. The coupler unit further includes a boss extending from the side plate. The coupler unit includes a slider configured to be movably coupled with the boss. Based on a contact of the work tool with the slider, the slider is configured to move relative to the boss from a first position to a second position. In the first position, the slider is misaligned relative to the boss and the side plate, and the slider prevents receipt of the pin within the boss and the side plate. In the second position, each of the coupling bracket, the slider, the boss, and the side plate are in alignment with each other, and the pin is receivable within each of the coupling bracket, the slider, the boss, and the side plate.

(8) In yet another aspect of the present disclosure, a method of coupling a work tool with a work machine is provided. The work tool includes a coupling bracket. The method includes moving a coupler unit towards the work tool. The coupler unit includes at least one side plate, a boss extending from the side plate, and a slider movably coupled with the boss. In a first position of the slider, the slider is misaligned relative to the boss and the side plate. The method also includes contacting the work tool with the slider. The method further includes moving the slider relative to the boss from the first position to a second position based on the contact of the work tool with the slider. In the second position of the slider, each of the coupling bracket, the slider, the boss, and the side plate are in alignment with each other.

(9) Other features and aspects of this disclosure will be apparent from the following description and the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic side view of a work machine, according to an embodiment of the present disclosure;
- (2) FIG. 2 is a schematic side view of a work tool and a coupler unit associated with the work machine of FIG. 1, according to an embodiment of the present disclosure;
- (3) FIG. 3 is a schematic perspective view of the coupler unit coupled with the work tool;
- (4) FIG. 4 is a schematic side view illustrating a slider of the coupler unit of FIG. 2 in a first position;
- (5) FIG. 5 is a schematic side view of the slider of FIG. 4;
- (6) FIG. 6 is a schematic top view of the coupler unit wherein the slider is in the first position;
- (7) FIG. 7 is a schematic side view of the slider in a second position, according to an embodiment of the present disclosure;
- (8) FIG. 8 is a schematic top view of the coupler unit coupled with the work tool when the slider is in the second position; and
- (9) FIG. 9 is a flowchart for a method of coupling the work tool with the work machine of FIG. 1, according to an embodiment of the present disclosure.

DETAILED DESCRIPTION

(10) Wherever possible, the same reference numbers will be used throughout the drawings to refer to the same or like parts. Referring to FIG. 1, a schematic side view of an exemplary work machine **100** is illustrated. The work machine **100** is embodied as a wheel loader that may be used for purposes, such as, construction, landscaping, agriculture, and the like. Alternatively, the work machine **100** may be embodied as an off highway truck, a dozer, an excavator, a track-type tractor, a motor grader, etc. that may be used in various industries to move, remove, or load materials, such as, asphalt, debris, dirt, snow, feed, gravel, logs, raw minerals, recycled material, rock, sand, woodchips, etc.

(11) The work machine **100** includes a frame **102**. The frame **102** supports various components of the work machine **100**. The work machine **100** defines a first side **104** (being visible, end on, in FIG. 1) and a second side opposite the first side **104**. Further, the work machine **100** includes an operator cabin **108** supported by the frame **102**. The operator cabin **108** may include one or more controls (not shown) that may enable an operator to control the work machine **100**. The work machine **100** further includes a hood **110** mounted on the frame **102**. Further, the work machine **100** includes a power source (not shown) supported by the frame **102**. The power source may be disposed within the hood **110**. The power source may include an engine, such as, an internal combustion engine, batteries, motors, and the like. The power source may provide power to various components of the work machine **100** for operational and mobility requirements.

(12) The work machine **100** includes a pair of front wheels **112** and a pair of rear wheels **114**. The front wheels **112** and the rear wheels **114** provide support and mobility to the work machine **100** on grounds. Alternatively, the work machine **100** may include tracks instead of the wheels **112**, **114**. The work machine **100** also includes a linkage assembly **116** coupled to the frame **102**. The linkage assembly **116** is movably coupled to the frame **102**. The linkage assembly **116** includes a pair of linkage arms **117** coupled to the frame **102** at one end thereof. The linkage assembly **116** also includes a coupler unit **120** (see FIGS. 2 and 3). The coupler unit **120** is omitted from FIG. 1 for clarity. The linkage assembly further includes a work tool **118**. The work tool **118** may be used to perform work operations, such as, loading, stock piling, dumping, and the like. The work tool **118**

is embodied as a bucket herein. Alternatively, the work tool **118** may be any other type of work tool known in the art, such as, a blade.

(13) FIG. **2** illustrates a schematic side view of the work tool **118**. While the work tool **118** is depicted in whole, as viewed from the first side **104** (same as FIG. **1**), a portion of the coupler unit **120**, nearest the viewer, is cutaway to illustrate parts that are facing away from the first side **104** (as described in more detail below). The work tool **118** includes a coupling bracket **122**. Specifically, the work tool **118** includes a pair of coupling brackets **122** (only one of which is illustrated herein). Each coupling bracket **122** defines a bore **123**. The work tool **118** also includes a stopper **121**. The stopper **121** may be fixedly coupled with the coupling bracket **122**. In an example, a material of the stopper **121** may be same as a material of the work tool **118**. The stopper **121** may be coupled to the coupling bracket **122** by welding, or any other fastening means, such as, a bolt, a screw, a rivet, and the like. The stopper **121** includes a flat profile **125**. It should be noted that a design of the stopper **121** as illustrated herein is exemplary in nature, and the stopper **121** may include any other design.

(14) Referring now to FIG. **3**, the coupler unit **120** couples with the work tool **118**. Specifically, the coupling brackets **122** of the work tool **118** allow coupling of the work tool **118** with the coupler unit **120**. The linkage arms **117** (see FIG. **1**) also couple with the coupler unit **120**. Thus, the work tool **118** is movably coupled to the frame **102** (see FIG. **1**), via the linkage arms **117** and the coupler unit **120**.

(15) As shown in FIG. **3**, the coupler unit **120** includes a pin **124**. The pin **124** couples with the coupling bracket **122** of the work tool **118**. In the illustrated example, the pin **124** is cylindrical in shape. In other examples, the pin **124** may have any other shape or design, based on application attributes. It should be noted that the coupler unit **120** includes two pins **124**. In some examples, the coupler unit **120** may include an indication system (not shown) that generates an alert or indication when the pin **124** is in a fully extended position.

(16) The coupler unit **120** also includes an actuator **126**. The pins **124** may extend/retract based on an actuation by the actuator **126**. The actuator **126** may include a hydraulic actuator, a pneumatic actuator, or an electrical actuator. In the illustrated example, the actuator **126** is embodied as a hydraulic actuator. The actuator **126** is in fluid communication with fluid lines **128** to extend/retract the pins **124** based on variation in fluid pressure.

(17) The coupler unit **120** further includes one or more side plates **130**. In the illustrated example, the coupler unit **120** includes two side plates **130**. Further, each side plate **130** includes a first opening **132** through which the pin **124** extends during the coupling of the work tool **118** with the coupler unit **120**.

(18) The coupler unit **120** also includes two primary plates **134** disposed proximate to a corresponding side plate **130**. Further, each primary plate **134** includes a second opening **140** through which the pin **124** extends during the coupling of the work tool **118** with the coupler unit **120**. It should be noted that, when the coupler unit **120** is to be coupled with the work tool **118**, each pin **124** extends so that the pin **124** couples with a corresponding primary plate **134**, a corresponding coupling bracket **122**, and a corresponding side plate **130**.

(19) The coupler unit **120** also includes a bar **142** that rigidly connects the side plate **130** and the primary plate **134**. The coupler unit **120** further includes a pair of secondary plates **144** having a pair of aligned openings **146**. The secondary plates **144** couple the coupler unit **120** with the linkage arms **117**.

(20) FIG. **4** illustrates a schematic side view of the coupler unit **120**. In FIG. **4**, the coupler unit **120** is viewed from the second side (opposite the point of view depicted in FIGS. **1** and **2**). The coupler unit **120** includes a boss **148** extending from the side plate **130**. Specifically, the boss **148** extends from an inner surface **150** of the side plate **130**. It should be noted that the coupler unit **120** may include two bosses **148** (only one of which is illustrated herein), each of which is coupled to a corresponding side plate **130**. The boss **148** defines a third opening **154**. The boss **148** also defines one or more slots **168**. Specifically, the boss **148** defines two slots **168** that are in alignment with

each other. The slots **168** are substantially linear and are disposed proximate to the third opening **154**.

(21) The coupler unit **120** also includes a slider **152** movably coupled with the boss **148**. It should be noted that the coupler unit **120** may include two sliders **152** (only one of which is illustrated herein), each of which is coupled to a corresponding boss **148**. In an example, a material of the slider **152** may be same as a material of the coupler unit **120**.

(22) Based on a contact of the work tool **118** (see FIGS. 2 and 3) with the slider **152**, the slider **152** moves relative to the boss **148** from a first position to a second position. The slider **152** is shown in the first position in FIG. 4. In FIG. 4, a movement of the slider **152**, from the first position to the second position, from the perspective of the viewer, is generally in a direction from left to right, whereas, in FIG. 2, which is reversed, a movement of the slider **152** is in the opposite direction, e.g., being generally from right to left. The slider **152** defines a central opening **160**. In the first position, the slider **152** is misaligned relative to the boss **148** and the side plate **130**, and the slider **152** prevents receipt of the pin **124** (see FIG. 3) within the boss **148** and the side plate **130**. In other words, in the first position of the slider **152**, the central opening **160** of the slider **152** is misaligned relative to the third opening **154** of the boss **148** and the first opening **132** (see FIG. 3) of the side plate **130**, thereby, preventing the extension of the pin **124** through the third opening **154** of the boss **148** and the first opening **132** of the side plate **130**. Further, the misalignment of the slider **152** relative to the boss **148** prevents coupling of the coupler unit **120** to the work tool **118**.

(23) FIG. 5 is a schematic side view of the slider **152**. The slider **152** includes a body portion **156**. The slider **152** also includes an engagement portion **158** extending from the body portion **156**. The work tool **118** (see FIGS. 2 and 3) contacts the engagement portion **158** for moving the slider **152** from the first position to the second position. In the illustrated example, the engagement portion **158** is substantially rectangular in shape. In other examples, the engagement portion **158** may have any other shape such as, a square shape.

(24) Referring now to FIGS. 4 and 5, the slider **152** includes one or more fasteners **166** to movably couple the slider **152** with the boss **148**. Specifically, the slider **152** includes a pair of fasteners **166** that movably couple the slider **152** with the boss **148**. The fasteners **166** are coupled with the body portion **156**. Further, the fasteners **166** are received within the one or more slots **168** defined in the boss **148**. Specifically, each fastener **166** is received within a corresponding slot **168**. The fasteners **166** may include, for example, a screw, a bolt, or any other securing means that allows coupling of the slider **152** with the boss **148**.

(25) Referring again to FIG. 4, the coupler unit **120** further includes a spring **164** coupled with and extending between the body portion **156** and the boss **148**. The spring **164** biases the slider **152** towards the first position. For example, the spring **164** moves the slider **152** to the first position when the work tool **118** is not in contact with the slider **152**. In other words, when the work tool **118** is not in contact with the slider **152**, the spring **164** moves the slider **152** to the first position so that the central opening **160** of the slider **152** is misaligned relative to the third opening **154** of the boss **148**.

(26) FIG. 6 is a schematic top view of the coupler unit **120** with the slider **152** in the first position. As stated earlier with respect to FIG. 4, and depicted from a different viewpoint in FIG. 6, in the first position of the slider **152**, the slider **152** is misaligned relative to the boss **148** and the side plate **130**. Thus, the slider **152** prevents receipt of the pin **124** (see FIG. 3) within the boss **148** and the side plate **130**.

(27) FIG. 7 is a schematic side view of the slider **152** in the second position. Referring now to FIG. 7, like in FIG. 2, the boss **148** is sectioned through and the side plate **130**, nearest the viewer, is cutaway to illustrate the boss **148** and the slider **152**, which are attached to the inner surface **150** of the side plate **130** that is facing away from the first side **104**. In FIG. 7, the work tool **118** contacts the engagement portion **158** for moving the slider **152** from the first position to the second position. Particularly, the work tool **118** includes the stopper **121** that contacts with the slider **152** for moving

the slider **152** from the first position to the second position. As illustrated in FIG. 7, the flat profile **125** of the stopper **121** contacts the engagement portion **158**, which causes the slider **152** to move to the second position. In the second position, each of the coupling bracket **122**, the slider **152**, the boss **148**, and the side plate **130** (see FIG. 4) are in alignment with each other. Further, the pin **124** (see FIG. 3) is receivable within each of the coupling bracket **122**, the slider **152**, the boss **148**, and the side plate **130**. Thus, when the stopper **121** of the work tool **118** pushes the slider **152** to the second position, the second opening **140** (see FIG. 3) of the primary plate **134** (see FIG. 3), the bore **123** of the coupling bracket **122** (also visible in FIG. 2), the central opening **160** of the slider **152**, the third opening **154** of the boss **148**, and the first opening **132** (see FIG. 4) of the side plate **130** are in alignment with each other.

(28) Referring now to FIG. 8, when the slider **152** is in the second position and the pin **124** (shown with dashed lines) is extended, the pin **124** may couple with each of the primary plate **134**, the coupling bracket **122**, the slider **152**, the boss **148**, and the side plate **130**, thereby coupling the coupler unit **120** with the work tool **118**.

(29) It should be noted the coupler unit **120** with the work tool **118** has a symmetric design. Accordingly, only the details of the coupler unit **120** with the work tool **118** disposed at the first side **104** (see FIG. 1) of the work machine **100** have been described herein. However, the details provided herein are equally applicable to the coupler unit **120** with the work tool **118** disposed at the second side of the work machine **100**, without any limitations thereto.

(30) It is to be understood that individual features shown or described for one embodiment may be combined with individual features shown or described for another embodiment. The above described implementation does not in any way limit the scope of the present disclosure. Therefore, it is to be understood although some features are shown or described to illustrate the use of the present disclosure in the context of functional segments, such features may be omitted from the scope of the present disclosure without departing from the spirit of the present disclosure as defined in the appended claims.

INDUSTRIAL APPLICABILITY

(31) The coupler unit **120** of the present disclosure allows coupling of various work tools with the work machine **100**. The coupler unit **120** includes the slider **152** movably coupled with the boss **148**. The slider **152** is movable between the first and second positions. The first position of the slider **152** may be indicative of a misalignment of the bore **123** of the coupling bracket **122** relative to the pin **124**. If the work tool **118** is not in contact with the slider **152**, the slider **152** may be in the first position and may prevent the extension of the pin **124** through the boss **148** and the side plate **130**. In other words, the slider **152** may prevent a full extension of the pin **124** when the work tool **118** is not in contact with the coupler unit **120**. Moreover, as the pin **124** does not extend fully due to the slider **152**, the pin **124** may not trigger the indication system of the coupler unit **120**, thereby preventing any generation of false coupling indications by the indication system.

(32) Further, when the coupler unit **120** is moved towards the work tool **118**, the stopper **121** of the work tool **118** contacts the slider **152** and pushes the slider **152** to the second position. Moreover, at such an instance, the bore **123** of the coupling bracket **122** aligns with the pin **124**. Thus, in the second position of the slider **152**, the bore **123** of the coupling bracket **122**, the central opening **160** of the slider **152**, the third opening **154** of the boss **148**, and the first opening **132** of the side plate **130** are in alignment with each other, and the pin **124** may extend therethrough thereby allowing coupling of the coupler unit **120** with the work tool **118**.

(33) The coupler unit **120** of the present disclosure may help in coupling the work tool **118** with the coupler unit **120** in lesser time. Since the slider **152** of the coupler unit **120** ensures, structurally and/or mechanically, the alignment of the pin **124** with the work tool **118**, the operator may not have to visually check for the misalignment, which may save operator efforts. The coupler unit **120** may be retrofitted on existing coupler units by performing minimal modifications on the coupler units. Further, the coupler unit **120** having the slider **152** may be reliable and may also exhibit

improved usability.

(34) Referring to FIG. 9, a flowchart for a method **900** of coupling the work tool **118** with the work machine **100** is illustrated. At step **902**, the coupler unit **120** is moved towards the work tool **118**. The work tool **118** includes the coupling bracket **122**. The coupler unit **120** includes the side plate **130**, the boss **148** extending from the side plate **130**, and the slider **152** movably coupled with the boss **148**. In the first position of the slider **152**, the slider **152** is misaligned relative to the boss **148** and the side plate **130**.

(35) At step **904**, the work tool **118** contacts with the slider **152**. The slider **152** includes the body portion **156** and the engagement portion **158** extending from the body portion **156**. Further, contacting the work tool **118** with the slider **152** includes contacting the stopper **121** of the work tool **118** with the engagement portion **158**.

(36) At step **906**, the slider **152** is moved relative to the boss **148** from the first position to the second position based on the contact of the work tool **118** with the slider **152**. In the second position of the slider **152**, each of the coupling bracket **122**, the slider **152**, the boss **148**, and the side plate **130** are in alignment with each other. Further, moving the slider **152** relative to the boss **148** also includes moving the fastener **166** coupled with the body portion **156** within the slot **168** defined in the boss **148**. Additionally, moving the slider **152** relative to the boss **148** includes retracting the spring **164** coupled with and extending between the body portion **156** and the boss **148**.

(37) Further, the pin **124** of the coupler unit **120** is extended based on the movement of the slider **152** from the first position to the second position. Moreover, the pin **124** is received within each of the coupling bracket **122**, the slider **152**, the boss **148**, and the side plate **130** for coupling the work tool **118** with the work machine **100**.

(38) Further, the slider **152** prevents the extension of the pin **124** when the work tool **118** is misaligned relative to the pin **124**. Moreover, when the work tool **118** is misaligned relative to the pin **124**, the generation of a false coupling indication of the engagement of the pin **124** and the work tool **118** is prevented based on the prevention of the extension of the pin **124**.

(39) While aspects of the present disclosure have been particularly shown and described with reference to the embodiments above, it will be understood by those skilled in the art that various additional embodiments may be contemplated by the modification of the disclosed work machine, systems and methods without departing from the spirit and scope of the disclosure. Such embodiments should be understood to fall within the scope of the present disclosure as determined based upon the claims and any equivalents thereof.

Claims

1. A coupler unit for coupling a work tool with a work machine, the work tool including a coupling bracket, the coupler unit comprising: a pin configured to couple with the coupling bracket of the work tool; at least one side plate; a boss extending from the side plate; and a slider movably coupled with the boss, the slider including an opening therethrough, wherein, based on a contact of the work tool with the slider, the slider is configured to move relative to the boss from a first position to a second position, wherein, in the first position, the opening of the slider is misaligned relative to the boss and the side plate, and the slider prevents receipt of the pin within the boss and the side plate, and wherein, in the second position, each of the coupling bracket, the opening of the slider, the boss, and the side plate are in alignment with each other, and the pin is receivable within each of the coupling bracket, the opening of the slider, the boss, and the side plate.
2. The coupler unit of claim 1, wherein the slider includes: a body portion; and an engagement portion extending from the body portion, wherein the work tool is configured to contact the engagement portion for moving the slider from the first position to the second position.
3. The coupler unit of claim 2, further comprising a spring coupled with and extending between the

body portion and the boss.

4. The coupler unit of claim 3, wherein the spring is configured to move the slider to the first position when the work tool is not in contact with the slider.

5. The coupler unit of claim 2, wherein the slider includes at least one fastener configured to movably couple the slider with the boss, and wherein the fastener is coupled with the body portion.

6. The coupler unit of claim 5, wherein the fastener is configured to be received within at least one linear slot defined in the boss and the slider slides along the linear slot.

7. The coupler unit of claim 1, wherein the work tool includes a stopper configured to contact with the slider for moving the slider from the first position to the second position.

8. A work machine comprising: a frame; and a linkage assembly coupled to the frame, the linkage assembly including: a work tool including a coupling bracket; and a coupler unit configured to couple with the work tool, the coupler unit including: a pin configured to couple with the coupling bracket of the work tool; at least one side plate; a boss extending from the side plate, the boss including a linear slot; and a slider slidably coupled to the slot of the boss, wherein, based on a contact of the work tool with the slider, the slider is configured to slide along the slot of the boss from a first position to a second position, wherein, in the first position, the slider is misaligned relative to the boss and the side plate, and the slider prevents receipt of the pin within the boss and the side plate, and wherein, in the second position, each of the coupling bracket, the slider, the boss, and the side plate are in alignment with each other, and the pin is receivable within each of the coupling bracket, the slider, the boss, and the side plate.

9. The work machine of claim 8, wherein the slider includes: a body portion; and an engagement portion extending from the body portion, wherein the work tool is configured to contact the engagement portion for moving the slider from the first position to the second position.

10. The work machine of claim 9, wherein the slider further includes a spring coupled with and extending between the body portion and the boss.

11. The work machine of claim 10, wherein the spring is configured to move the slider to the first position when the work tool is not in contact with the slider.

12. The work machine of claim 8, wherein the slider includes an opening therethrough, and wherein in the second position, the coupling bracket, the opening of the slider, the boss, and the side plate are in alignment with each other, and the pin is receivable within each of the coupling bracket, the opening of the slider, the boss, and the side plate.

13. The work machine of claim 12, wherein in the second position, the pin passes through the opening of the slider.

14. The work machine of claim 8, wherein the work tool includes a stopper configured to contact with the slider for moving the slider from the first position to the second position.

15. A method of coupling a work tool with a work machine, the work tool including a coupling bracket, the method comprising: moving a coupler unit towards the work tool, the coupler unit including at least one side plate, a boss extending from the side plate, and a slider movably coupled with the boss, the slider including an opening therethrough, wherein, in a first position of the slider, the opening of the slider is misaligned relative to the boss and the side plate; contacting the work tool with the slider; and moving the slider relative to the boss from the first position to a second position based on the contact of the work tool with the slider, wherein, in the second position of the slider, each of the coupling bracket, the opening of the slider, the boss, and the side plate are in alignment with each other.

16. The method of claim 15, wherein the slider includes a body portion and an engagement portion extending from the body portion, and wherein contacting the work tool with the slider further includes contacting a stopper of the work tool with the engagement portion.

17. The method of claim 16, wherein moving the slider relative to the boss further includes moving at least one fastener coupled with the body portion within at least one slot defined in the boss.

18. The method of claim 16, wherein moving the slider relative to the boss further includes

retracting a spring coupled with and extending between the body portion and the boss.

19. The method of claim 15 further comprising: extending a pin of the coupler unit based on the movement of the slider from the first position to a second position; and receiving the pin within each of the coupling bracket, the opening of the slider, the boss, and the side plate for coupling the work tool with the work machine.

20. The method of claim 19, further comprising: preventing, by the slider, an extension of the pin when the work tool is misaligned relative to the pin; and preventing, when the work tool is misaligned relative to the pin, a generation of a false coupling indication of an engagement of the pin and the work tool based on the prevention of the extension of the pin.
